

performance. FDMA uses a frequency canalisation approach to spectrum management, while time-division multiple access (TDMA) utilises a time-division approach. The entire available cellular RF spectrum is sub-divided into narrow-band radio channels to be used as a one-way communication link between cellular mobile units and base stations.

Multiple access technologies for cellular systems

Generally, a fixed amount of frequency spectrum is allocated to a cellular system. Multiple access techniques are deployed so that the users can share the available spectrum in an efficient manner.

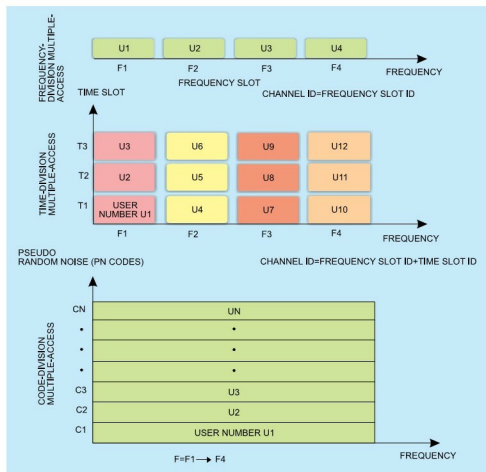
For wireless communication, multiplexing can be carried out in three dimensions: Time (TDMA), frequency (FDMA and its variation OFDMA) and code (CDMA).

In TDMA the available spectrum is partitioned into narrow frequency bands or frequency channels, which, in turn, are divided into a number of time slots. In case of North American digital cellular standard IS-136, each frequency channel (30 kHz) is divided into three time slots, whereas in European digital cellular system GSM each frequency channel (200 kHz) is divided into eight time slots. Guard bands are needed both between frequency channels and time slots.

In FDMA, users share the available spectrum in a frequency band called traffic channel. Different users are assigned different channels on demand basis. The user's signal power is concentrated in a relatively narrow frequency band. All the analogue cellular systems used FDMA system.

OFDM is a multi-cellular transmission technique where a data stream is carried with many lower-rate sub-carrier tones. It has been adopted in mobile communications to combat hostile frequency-selective fading and has been incorporated into wireless network standards.

OFDM combines the advantages of coherent detection and OFDM modulation and has many merits that are criti-



FDMA, TDMA, and CDMA multiple-access techniques

cal for future high-speed transmission systems. By using up/down conversion, electrical bandwidth requirement can be greatly reduced for the OFDM transceiver, which is extremely attractive for high-speed circuit design where electrical signal bandwidth dictates the cost. Lastly, signal processing in the OFDM transceiver can take advantage of efficient algorithm of fast Fourier transform (FFT)/inverse FFT, which suggests that OFDM has superior scalability over channel dispersion and data rate.

Digital modulation keying

Communication systems often involve modulation of a carrier, which results in a bandpass waveform. A digital signal can be used to modulate the amplitude, frequency or phase of a sinusoidal carrier producing three different forms of digital modulation: amplitude-shift keying (ASK), frequency-shift keying (FSK) and phase-shift keying (PSK). In addition to these basic techniques, there are some modulation

schemes that employ a combination of amplitude and phase modulation. It may be noted that unlike ASK signal, PSK transmission is polar. At the same time, ASK is a linear modulation scheme, whereas PSK is a non-linear modulation scheme. PSK has a superior performance over ASK.

Quadrature phase-shift keying (QPSK). Digital modulation techniques mentioned above are spectrally inefficient in the sense that the available channel bandwidth is not fully used. Spectral efficiency can be improved by using QPSK. It is a system for two message sources. In this system modulation carriers in phase quadrature are combined to form the output waveform. In QPSK the amplitude of the modulator waveform and modulator gains are made as nearly equal as possible.

Differential phase-shift keying (DPSK). DPSK is a modification of PSK that avoids the need to provide synchronous carrier required for detection of PSK signals. It is an ingenious technique whereby the carrier reference is